MCQ

Group-1:

ID:04:

1. Which of the following is not a valid string declaration in C++?

a. char str[] = "Hello World";

b. char *str = "Hello World";

c. string str = "Hello World";

d. wchar_t str[] = L"Hello World";

e. All of the above are valid.

Answer: b

2. What is the output of the following code?

string str = "Hello World";

cout << str[6];

a. H

b. e

c. W

d. o

e. None of the above

Answer: W

3. Which of the following is not a valid function overloading in C++?

a. int add(int a, int b);

b. double add(double a, double b);

c. int add(int a, double b);

d. void add(int a, int b);

e. All of the above are valid.

4. What is function overloading in C++?

a. Defining multiple functions with the same name but different parameters.

b. Defining multiple functions with the same name but different return types.

c. Defining multiple functions with the same name and same parameters.

d. Defining multiple functions with different names and same parameters.

e. None of the above.

Ans: a

5. What is the default access specifier in a C++ class?

a. public

b. private

c. protected

d. friend

e. None of the above.

Answer: private

6. Which of the following is not a valid access specifier in C++?

a. public

b. private

c. protected

d. friend

e. All of the above are valid.

Answer: e

7. Which of the following is not a member function of the string class in C++?

A. append()

B. length()

C. delete()

D. substr()

E. find()

Solution: C. delete()

8. Which C++ library provides the string class?

A. iostream

B. string.h

C. stdio.h

D. string

E. cstring

Solution: D. string

ID:05

1. Who is credited with developing the concept of OOP?

a) Bill Gates b) Steve Jobs c) Alan Turing d) Alan Kay e)None of them

Answer: d) Alan Kay

2. Which of the following is not a key concept of OOP?

a) Abstraction b) Inheritance c) Encapsulation d) Linear Programming e) All level programming

Answer: d) Linear Programming

3. What is the purpose of polymorphism in OOP?

a) To enable objects of different classes to be treated as if they were objects of the same class

b) To hide the implementation details of an object and only expose necessary methods and properties

c) To enable objects to access methods and properties of another object

d) To enable objects to send and receive messages to and from other objects

e) To design model.

Answer: a) To enable objects of different classes to be treated as if they were objects of the same class.

4. Can you overload a function based on the const-ness of the object passed to it?

a) Yes

b) No

c) It depends on the function implementation

d) It depends on the parameter type

e) It depends on the return type

Answer: a) Yes

5. Which of the following is an example of function overloading?

a) int add(int a, int b)

b) double add(double a, double b)

c) void add(int a, int b)

d) All of the above

e) None of the above

Answer: d) All of the above

6. Which of the following is a valid way to initialize a string in C++?

a) std::string s = "hello";

b) std::string s("hello");

c) std::string s = {'h', 'e', 'l', 'l', 'o'};

d) All of the above

e) None of the above

Answer: d) All of the above

7. Which of the following is a valid way to access the first character of a string in C++?

a) s[0]

b) s.front()

c) *s.begin()

d) All of the above

e) None of the above

Answer: d) All of the above

8. What is the difference between the length() and size() functions of a string object in C++?

a) There is no difference, they both return the number of characters in the string

b) length() returns the number of characters in the string, while size() returns the number of bytes used by the string

c) length() is a member function of the string class, while size() is a global function

d) size() returns the number of characters in the string, while length() returns the number of bytes used by the string

e) length() is faster than size() because it does not have to count the null terminator

Answer: a) There is no difference, they both return the number of characters in the string

____

ID:06

1. Which of the following is NOT one of the three philosophies that make up OPP? a) Objectivism b) Peacism c) Progressivism d) Utilitarianism

Answer: d) Utilitarianism

2. According to Objectivism, what is the primary means of understanding reality?

a) Emotion b) Faith c) Reason d) Intuition

Answer: c) Reason

3. Which of the following is a key value emphasized by Peacism?

a) Individualism b) Rational self-interest c) Nonviolence d) Social progress

Answer: c) Nonviolence

4. Which of the following civilizations emerged in Mesopotamia around 4000 BCE? a) Egyptian b) Greek c) Roman d) Sumerian

Answer: d) Sumerian

5. Which of the following events marked the end of the Roman Republic and the beginning of the Roman Empire?

a) The assassination of Julius Caesar b) The Battle of Actium c) The reign of Augustus Caesar d) The sack of Rome by the Visigoths

Answer: c) The reign of Augustus Caesar

6. Which of the following is a benefit of function overloading?

a) It allows for more efficient code execution b) It simplifies the code by reducing the number of functions c) It improves code readability and maintainability d) It ensures that functions can only be called with the correct number of parameters

Answer: c) It improves code readability and maintainability

7. Which of the following is an example of function overloading?

a) int add(int x, int y) b) double add(double x, double y) c) void add(int x, int y) d) All of the above

Answer: d) All of the above

8. Which of the following is NOT a way to overload a function in C++20?

a) By changing the number of parameters b) By changing the return type c) By changing the order of the parameters d) By changing the scope of the function

Answer: d) By changing the scope of the function

ID:0622220105101007

Topic: Philosophy of OOP, Function overloading, String

Philosophy of OOP:

1. What is C++?

a) C++ is an object oriented programming language

b) C++ is a procedural programming language

c) C++ supports both procedural and object oriented programming language

d) C++ is a functional programming language

Answer: c

2. Which of the following user-defined header file extension used in c++?

a) .hg

b) .cpp

c) .h

d). hf

Answer: c

3. What will be the output of the following C++ code?

```
#include <iostream>
#include <string>
using namespace std;
int main(int argc, char const *argv[])
{
	char s1[6] = "Hello";
	char s2[6] = "World";
	char s3[12] = s1 + " " + s2;
	cout<<s3;
	return 0;
}
```

a) Hello

b) World

c) Error

d) Hello World

Answer: c

Function Overloading:

1. Which of the following permits function overloading on c++?

a) type

b) number of arguments

c) type & number of arguments

d) number of objects

Answer: c

2. In which of the following we cannot overload the function?

a) return function

b) caller

c) called function

d) main function

Answer: a

3. Function overloading is also similar to which of the following?

a) operator overloading

b) constructor overloading

c) destructor overloading

d) function overloading

Answer: b

String:

1. What is string objects in C++?

a) Stream of alphabets

b) A stream of well-defined characters

c) Stream of characters

d) A stream of characters terminated by \0

Answer: b

2. What will be the output of the following C++ code?

```
#include <iostream>
#include <string>
using namespace std;
int main(int argc, char const *argv[])
{
	char str[] = "Hello World";
	cout<<str[0];
	return 0;
}
```

a) H

b) e

c) Error

d) o

Answer: a

15. What is the identifier given to string class to declare string objects?

a) String

b) string

c) STRING

d) Any of the above can be used

Answer: b

____

ID:39

1.One of the basic concepts in object Oriented Programming approach is bundling both data and functions into one unit known as-

a.simple variable

b.object

c.Inheritance

d.Bundle

Ans: b

2. Overloaded functions in C++ oops are

a.Functions preceding with virtual keywords.

b.Functions inherited from base class to derived class.

c.Two or more functions having same name but different number of parameters or type.

d.None of above

Ans: b

3.Who invented OOP?

a) Alan Kay

b) Andrea Ferro

c) Dennis Ritchie

d) Adele Goldberg

Ans: a

4. Output?

```
#include<iostream>
using namespace std;
int fun(int x = 0, int y = 0, int z)
{
return (x + y + z);
}
int main()
{
cout << fun(10);
return 0;
}
```

a. 0

b. 10

c. 20

d. Compiler Error

Ans- d

5.Function overloading is also similar to which of the following?

a. function overloading

b. destructor overloading

c. operator overloading

d. constructor overloading

Ans :- d

6. What will be the output of the following C++ code?

```
#include <iostream>

#include <string>

using namespace std;

int main(int argc, char const *argv[])

{

        string s1 = "Hello";

        string s2 = "World";

        string s3 = s1 + " " + s2;

        cout<<s3;

        return 0;

}
```

a.Hello World

b. Hello

c. World

d. Error

Ans: a

7. Which of the following is not a modifier function in string class?

a. operator+=()

b. operator[]()

c. push_back()

d. erase()

Ans: b

8. Which of the following language supports polymorphism but not the classes?

a.C++ programming language

b.Java programming language

c.Ada programming language

d.C# programming language

Ans: c

ID: 0622220105101010

Topics: Philosophy of OOP, Function Overloading, String

- Philosophy of OOP:

1. Which feature of OOP indicates code reusability?

   a) Abstraction

   b) Polymorphism

   c) Encapsulation

   d) Inheritance

Answer: d

2. How many types of access specifiers are provided in OOP (C++)?

   a) 4

   b) 3

   c) 2

   d) 1

Answer: b

3. Which constructor will be called from the object created in the below C++ code?

```
class A
{
        int i;
        A()
        {
                i=0; cout&lt;&lt;i;
        }
        A(int x=0)
        {
                i=x;  cout&lt;&lt;I;
        }
```

};

A obj1;

a) Parameterized constructor

b) Default constructor

c) Run time error

d) Compile time error

Answer: d

- Function Overloading:

1. Which of the following permits function overloading on c++?

a) type

b) number of arguments

c) type & number of arguments

d) number of objects

Answer: c

2. What should be passed in parameters when function does not require any parameters?

a) void

b) blank space

c) both void & blank space

d) tab space

Answer: b

3. Overloaded functions are _________________

a) Very long functions that can hardly run

b) One function containing another one or more functions inside it

c) Two or more functions with the same name but different number of parameters or type

d) Very long functions

Answer: c

- String:

1. What is Character-Array?

a) array of alphabets

b) array of well-defined characters

c) array of characters

d) array of characters terminated by \0

Answer: c

2. What will be the output of the following C++ code?

```
#include <iostream>
#include <string>
using namespace std;
int main(int argc, char const *argv[])
{
        string str;
        cin>>str;
        cout<<str;
        return 0;
}
```

a) str

b) Input provided by the user

c) Error

d) Segmentation fault

Answer: b

3. Which header file is used to include the string object functions in C++?

a) #include <string.h>

b) #include <cstring>

c) #include <string>

d) #include <string.cpp>

Answer: c

GROUP-2:

ID : 0622220105101011

Topic : Advantages of OOP over Structured Programming , Inline function , Friend function.

1.Can a friend function be inherited by a derived class?

a. Yes

b.No

c. Sometimes

d.It depends on the specific use case

e.None of the above

2.What is a friend function inC++?

a.A function that is use to access private data members of a class called from within the same class

b.A function that is defined

c.A function that can onlu be outside the class,but has access to the private and protected members of the class

d.A function that is declared using the "friend" keyword in the class definition

e.None of the above

3.Can a friend function be a member function of another class?

a.Yes

b.No

c. Sometimes

d.It depends on the specific use case

e. None of the above

4.Can a recursive function be inline in C++?

a.Yes

b.No

c. Sometimes

d.It depends on the specific use case

e.None of the above

5.What is an inline function in C++?

a.A function that is only called once in a program

b.A function that is defined in the header file and can be directly subtituted into the calling code

c.A function that has a return type of void

d.A function that can onlu be called from within the same class

e.None of the above

6.What is the advantage of using an inline function?

a.It reduces the size of the program

b.It improves the speed of the program

c.It makes the code more readable

d.It allows the function to be called from anywhere in the program

e.It improves the security of the program

7.Can a virtual function be inline in C++?

a.Yes

b.No

c. Sometimes

d. It depends on the specific use case

e.None of the above

8.How does a friend function different from a member function?

a.A friend function can access private members of the class,while a member function cannot b.A member function can access private members of the class, while a friend function cannot

c.A friend function can be called from anywhere in the program,while a member function can

only be called from within the class

d.A member function can be declared using the "friend" keyword,while a friend function cannot

e.None of the above

ID : 0622220105101012

Topic : Advantages of OOP over Structured Programming , Inline function , Friend

function.

1.The time at which the executable code is started running is called?

(A). Critical time

(B). Compile time

(c). run time

(D). None of these

2.What is the effect of the Inline function on executable size?

(A). No effect on executable size

(B). decreases the executable size

(c). increases the executable size

(D). None of these

3.What is the effect of Inline Function on the locality of reference by

utilizing instruction cache?

(A). No effect

(B). decreases

(c). increases

(D). None of these

4. If class A is a friend of B, then B doesn't become a friend of A automatically.

A. TRUE

B. FALSE

C. Can be true and false

D. Can not say

5. Which keyword is used to represent a friend function?

A. friend

B. Friend

C. friend_func

D. Friend_func

6. Which of the following is correct about friend functions?

A. Friend functions use the dot operator to access members of a class using class objects

B. Friend functions can be private or public

C. Friend cannot access the members of the class directly

D. All of the above

7. Where does keyword 'friend' should be placed?

A. function declaration

B. function definition

C. main function

D. block function

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Topic : Advantages of OOP over Structured Programming , Inline function , Friend function.

1. Which of the following is an advantage of OOP over structured programming?

a. Encapsulatoin

b. Go to statement

c. Break statment

d. Continue statement

e. Switch statement

Answer : a

2.Which of the following is a feature of OOP that enables the impementation of polymorphism?

a. Inheeritan

b. Encapsulation

c. Abstruction

d. Modularity

e. Data hiding

Answer : a

3.Which of the following is a techique used in OOP to bind data and function together?

a.Polymorphisom

b. Encapsulation

c.Inheritance

d.Abstruction

e.Composition

Answer : b

4.Which of the following is a characteristic of an inline function in c++ ?

a. It is a separated function

b. It is defined outside the class

c. It is called explicity by the prigram

d.It is called implicity by the program

e.It is used to overload operators

Answer : d

5. Which of the following is an advance of ising inline function?

a.Increased program speed

b.decreased program speed

c. Increased program size

d. Decreased program size

e. Increased memory usage

Answer : a

6.Which of the following is an advantage of using friend function in c++?

a.Static function

b.Virtual function

c.Inline function

d.Member function

e. External function

Answer : d

7.Which of the following is an advantage od using friend function in c++ ?

a.Increased prorgam speed

b.Decreased program speed

c.Increased program size

d.Decreased program size

e.Impoved encapsulation

Answer : e

8. Which of the following is a techique used in OOP to allow one class to inherit

propertices and mehods from another class ?

a. Polymorphism

b. Encapsulation

c.Inheritance

d. Abstruction

e.Composition

Answer : c

9.Which of the following is a characteristic of a friend function in c++ ?

a. It is a member function of a class

b. It is declared outside the class

c. It is called explicitly by the program

d.It is called implicity by the program

e. It is used to overload operators

Answer : b

ID : 0622220205101014

Topic 1 : Advantages of OOP over structured programming.

1.which is not a feature of oop in general definitions?

A. Efficient code.

B. code reusability.

c. Modularity.

D. Duplicate/redundant data.

Answer: D

2. which feature of oop indicates code?

A. Abstruction.

B. polymorphism.

c. Encapsulation.

D. inheritance.

Answer: D

Topic 2: Inline function.

3.What is the effect on excutable size?

A. No effect on executable size.

B. decreases the executable size.

C. increases the executable size.

D. none of these.

Answer: C

4.What is the effect of inline function on memory?

A. can cause thrashing.

B. can cause more number of page faults

C. Both a and b.

D. none of these.

Answer: C

5. Request for making a function inline can be……..?

A. Accepted by the compiler.

B. Rejected by the compiler.

C. can be accepted or rejected depends on function size.

D. none of these.

Answer: C

Topic 3: Friend function.

6. A friend function can be used to…..

A. avoid arguments between classes.

B. allow one class to access an unrelated class.

C. allow access to classes whose source code is unavailable.

D.none of the avove.

Answer: B

7.The protocols for friend functions must be listed in the….

A. public declaration section.

B. privet declaration section.

C. implementation section.

D. none of these.

Answer: D

8. A friend function to a class , C cannot access…

A.protected data members.

B. the data members of the derived class of c.

C. private data members and member functions.

D. public data members and member function.

Answer: B

ID : 0622220205101015

Topic : Advantages of OOP over Structured Programming , Inline function , Friend function.

Advantages of OOP over Structured Programming

1. Which feature of OOP indicates code reusability?

a) Abstraction

b) Polymorphism

c) Encapsulation

d) Inheritance

Answer: d

2. Which header file is required in C++ to use OOP?

a) OOP can be used without using any header file

b) stdlib.h

c) iostream.h

d) stdio.h

Answer: a

3.Which of the following is not an OOPS concept?

a.Encapsulation

b.Polymorphism

c.Exception

d.Abstraction

Answer: c. Exception

Inline function

4. An inline function is expanded during _______________

a) compile-time

b) run-time

c) never expanded

d) end of the program

Answer: a

5. What is syntax of inline function?

(A). return-type

inline function-name(parameters)

{// Your code}

(B). return-type

function-name(parameters) inline {// Your code}

(c). inline return-type

function-name(parameters)

{// Your code}

(D). None of these

Answer : c

6. When the inline function is resolved/expanded?

(A). compile time

(B). run time

(c). Before run time

(D). circle time

Answer : a

7. What is an inline function?

a) A function that is expanded at each call during execution

b) A function that is called during compile time

c) A function that is not checked for syntax errors

d) A function that is not checked for semantic analysis

Answer: A

Friend function

8. Which keyword is used to represent a friend function?

a) friend

b) Friend

c) friend_func

d) Friend_func

Answer: a

9. Where does keyword ‘friend’ should be placed?

a) function declaration

b) function definition

c) main function

d) block function

Answer: a

10. What is the syntax of friend function?

a) friend class1 Class2;

b) friend class;

c) friend class

d) friend class()

Answer: a

ID: 0622220205101016

Group 2: Advantages of OOP over structural programming. Inline function, friend function.

MCQ:

1.Out of the following, which is not a member of the class?

a. Static function

b. Friend function

c. Constant function

d. Virtual function

ans: b

2.What is the other name used for functions inside a class?

a. Member variables

b. Member functions

c. Class functions

d. Class members

ans: b

3. Which of the following cannot be a friend?

a. Function

b. Class

c. Object

d. Operator function

ans: c

4. If a program usage Inline function ,then the function is expanded inline at

______________.

a. Compile time

b. Run time

c. Both a &b

d. None of these

ans: a

5. What is the effect of Inline function on executable size?

a. No effect on executable size

b. Decreases the executable size

c. Increases the executable size

d. None of these

ans: c

6. Which of the followings is/are true about friend function?

a. it can be called with class object

b. it can have objects as arguments

c. it can have built-in types as arguments

d. it doesn’t have this pointer as an argument

ans: b

7. Where does keyword “friend” should be placed?

a. function declaration

b. function definition

c. main function

d. block function

ans: a

8. What is an inline function ?

A. A function that is expanded at each all during execution

b. A function that is called during compile time

c. A function that is not checked for syntax errors

d. A function that is not checked for semantic analysis

ans: a

Group 3

Topic : Object, class, static non-static members

1. **Which of the following is not correct for virtual function in C++ ?.**

A. Virtual function can be static. C. Virtual function is defined in base class

B. Virtual function should be accessed using pointers D. Must be declared in public section of class

Ans : A

2. **How can we make a class abstract?**

A. By declaring it abstract using the static keyword. C. By making at least one member function as pure virtual B. By declaring it abstract using the virtual keyword. function

D. By making all member functions constant

Ans : C

3. **How many specifiers are present in access specifiers in class?**

A. 2 C. 4

B. 1 D. 3

Ans : D

4. **Which is also called as abstract class?**

A. Virtual function C. Pure virtual function

B. Derived class D. None of the mentioned

Answer: C

5. **If we stored five elements or data items in an array, what will be the index address or the index number of the array's last data item?**

A. 3 B. 5

C. 4 D. 88

Answer : C

6. **Which of the following is not correct for virtual function in C++ ?.**
A. Virtual function can be static. C. Virtual function is defined in base

class.
B. Virtual function should be accessed using pointers.   D. Must be declared in public section of class.
Answer : A

7. **How can we make a class abstract?**

A. By declaring it abstract using the static keyword   C. By making at least one member function as pure virtual
B. By declaring it abstract using the virtual keyword.   function
D. By making all member functions constant

Answer : C

8. How many specifiers are present in access specifiers in class?

A. 2   C. 4
B. 1   D. 3

Answer: D

9.Whenever any static data member is declared in a class ______________________

a) Only one copy of the data is created

b) New copy for each object is created

c) New memory location is allocated with each object

d) Only one object uses the static data

Ans: a

10. If object of class are created, then the static data members can be accessed ____________

a) Using dot operator

b) Using arrow operator

c) Using colon

d) Using dot or arrow operator

Ans: d

11. The static members are ______________________

a) Created with each new object

b) Created twice in a program

c) Created as many times a class is used

d) Created and initialized only once

Ans: d

12. The keyword static is used _______________

a) With declaration inside class and with definition outside the class

b) With declaration inside class and not with definition outside the class

c) With declaration and definition wherever done

d) With each call to the member function

Ans: b

13. The static member functions ____________________

a) Can be called using class name

b) Can be called using program name

c) Can be called directly

d) Can't be called outside the function

Ans: a

14. The static data member _________________

a) Can be mutable

b) Can't be mutable

c) Can't be integer

d) Can't be characters

Ans: b

15. Objects can be used _____________________

a) To access any member of a class

b) To access only public members of a class

c) To access only protected members of a class

d) To access only private members of a class

Ans: b

16. Which object can be used to access the standard input?

a) System.inner

b) cin

c) System.stdin

d) console.input

Ans: b

17. Which of the following operator is used to access static members of the class?

A) .

B) :

C) ::

D) ->

Answer : C

18. What is the correct output of given code snippets?

```
#include <iostream>
using namespace std;

class Sample {
    static int A;

public:
    static void set()
    {
        A = 20;
    }
    static void print()
    {
        cout << A << endl;
    }
};

int Sample::A = 10;

int main()
{
    Sample.set();
```

```
    Sample.print();
    return 0;
}
```

Options:

```
A.  10

B.  20

C.  Compile-time error

D.  Runtime error
```

```
Answer: 3. Compile-time error
```

19. **Which of the following is correct about static variables?**
a) Static functions do not support polymorphism
b) Static data members cannot be accessed by non-static member functions
c) Static data members functions can access only static data members
d) Static data members functions can access both static and non-static data members

Answer: C

20. What will be the output of the following C++ code?

```
#include <stdio.h>
int main()
{
   const int x;
   x = 10;
   printf("%d", x);
   return 0;
}
```

A. 10
B.Garbage value
C.Error
D. Segmentation fault

Answer: C

21. What will be the output of the following C++ code?

```
#include <iostream>
int const s=9;
int main()
```

```
{
    std::cout << s;
    return 0;
}
```

a) 9
b) Garbage value
c) Error
d) Segmentation fault

Answer: A

22. Which of the following is true?

A. Static methods cannot be overloaded.
B. Static data members can only be accessed by static methods.
C. Non-static data members can be accessed by static methods.
D. Static methods can only access static members (data and methods)
Answer: D

23.Output of following C++ program?

```
#include <iostream>

class Test

{

public:

    void fun();

};

static void Test::fun()

{

    std::cout<<"fun() is static\n";

}

int main()

{

    Test::fun();

    return 0;

}
```

Contributed by **Pravasi Meet**
A. fun() is static
B. Empty Screen
C. Compiler Error
D. No Output

Answer: C

24. Which of the following class allows to declare only one object of it?
a) Abstract class
b) Virtual class
c) Singleton class
d) Friend class

Answer: C

25. Which among the following is not applicable for the static member functions?

a) Variable pointers

b) void pointers

c) this pointer

d) Function pointers e) None of these

Answer: (C)

26. The static member functions ____________________

a) Can be called using class name

b) Can be called using program name

c) Can be called directly

d) Can't be called outside the function e) Options a and b

Answer: (A)

27. Which is correct syntax to access the static member functions with class name?

a) className . functionName;

b) className -> functionName;

c) className : functionName;

d) className :: functionName; e) None of these

Answer: (D)

28. Which keyword should be used to declare static variables?

a) static

b) stat

c) common

d) const e) all of them

Answer: (A)

29. The static data member ______________________

a) Must be defined inside the class

b) Must be defined outside the class

c) Must be defined in main function

d) Must be defined using constructor e) Options a and b

Answer: (B)

30. Which data members among the following are static by default?

a) extern

b) integer

c) const

d) void e) all of them

Answer: (C)

31. Which among the following can’t be used to access the members in any way?

a) Scope resolution

b) Arrow operator

c) Single colon

d) Dot operator e) None of these

Answer: (C)

32. We can use the static member functions and static data member __________________

a) Even if class object is not created

b) Even if class is not defined

c) Even if class doesn’t contain any static member

d) Even if class doesn’t have complete definition e) None of these

Answer: (A)

33.Which of the following is not a type of Constructor?
a) Friend constructor
b) Copy constructor
c) Default constructor
d) Parameterized constructor

Answer: **A**

34. Which of the following is correct?
a) Base class pointer object cannot point to a derived class object
b) Derived class pointer object cannot point to a base class object
c) A derived class cannot have pointer objects
d) A base class cannot have pointer objects

Answer: **B**

35.Out of the following, which is not a member of the class?
a) Static function
b) Friend function
c) Constant function
d) Virtual function

Answer: **B**

36. What is the other name used for functions inside a class?
a) Member variables
b) Member functions
c) Class functions

d) Class variables
Answer: **B**

37. Which of the following cannot be a friend?
a) Function
b) Class
c) Object
d) Operator function

Answer: **C**

38. Why references are different from pointers?
a) A reference cannot be made null
b) A reference cannot be changed once initialized
c) No extra operator is needed for dereferencing of a reference
d) All of the mentioned

Answer: **B**

39. Which of the following provides a programmer with the facility of using object of a class inside other classes?
a) Inheritance
b) Composition
c) Abstraction
d) Encapsulation

Answer: **B**

40. How run-time polymorphisms are implemented in C++?
a) Using Inheritance
b) Using Virtual functions
c) Using Templates
d) Using Inheritance and Virtual functions

Answer: **D**

41. How compile-time polymorphisms are implemented in C++?
a) Using Inheritance
b) Using Virtual functions
c) Using Templates
d) Using Inheritance and Virtual functions

Answer: **C**

42.Which of the following is an abstract data type?
a) int
b) float
c) class
d) string

Answer: **C**

43.Which concept means the addition of new components to a program as it runs?
a) Data hiding
b) Dynamic binding
c) Dynamic loading
d) Dynamic typing

Answer: **C**

44. Which of the following approach is used by C++?
a) Top-down
b) Bottom-up
c) Left-right
d) Right-left

Answer: **B**

45. Which operator is overloaded for a cout object?
a) >>
b) <<
c) <
d) >

Answer: **B**

46. Which of the following cannot be used with the virtual keyword?
a) Class
b) Member functions
c) Constructors
d) Destructors

Answer: **C**

47. Which of the following is correct?
a) C++ allows static type checking
b) C++ allows dynamic type checking.
c) C++ allows static member function to be of type const.
d) C++ allows both static and dynamic type checking

Answer: **D**

48. Which concept is used to implement late binding?
a) Virtual functions
b) Operator functions
c) Constant functions
d) Static functions

Answer: **A**

Group 5

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Abdus Salam

1. Which is the pointer which denotes the object calling the member function?

   a) Variable pointer

   b) This pointer

   c) Null pointer

   d) Zero pointer

   Answer: b

2. The this pointer is accessible ___________________

   a) Within all the member functions of the class

   b) Only within functions returning void

   c) Only within non-static functions

   d) Within the member functions with zero arguments

   Answer: c

3. Which is the correct interpretation of the member function call from an object, object.function(parameter);

   a) object.function(&this, parameter)

   b) object(&function,parameter)

   c) function(&object,&parameter)

   d) function(&object,parameter)

Answer: d

4. The address of the object _________________

a) Can’t be accessed from inside the function

b) Can’t be accessed in the program

c) Is available inside the member function using this pointer

d) Can be accessed using the object name inside the member function

Answer: c

5. The this pointers _____________________

a) Are modifiable

b) Can be assigned any value

c) Are made variables

d) Are non-modifiable

6. Which syntax doesn’t execute/is false when executed?

a) if(&object != this)

b) if(&function !=object)

c) this.if(!this)

d) this.function(!this)

Answer: a

7. This pointer can be used directly to ___________

a) To manipulate self-referential data structures

b) To manipulate any reference to pointers to member functions

c) To manipulate class references

d) To manipulate and disable any use of pointers

Answer: a

8. Which among the following is/are type(s) of this pointer?

a) const

b) volatile

c) const or volatile

d) int

Answer: c

9. Which is the correct syntax for declaring the type of this in a member function?

a) classType [cv-qualifier-list] *const this;

b) classType const[cv-qualifier-list] *this;

c) [cv-qualifier-list]*const classType this;

d) [cv-qualifier-list] classType *const this;

Answer: d

10. An object’s this pointer ______________________

a) Isn’t part of class

b) Isn’t part of program

c) Isn’t part of compiler

d) Isn’t part of object itself

Answer: d

11. The result of sizeof() function ___________________

a) Includes space reserved for this pointer

b) Includes space taken up by the address pointer by this pointer

c) Doesn’t include the space taken by this pointer

d) Doesn’t include space for any data member

Answer: c

12. Which among the following is true?

a) this pointer is passed implicitly when member functions are called

b) this pointer is passed explicitly when member functions are called

c) this pointer is passed with help of pointer member functions are called

d) this pointer is passed with help of void pointer member functions are called

Answer: a

Access Specifier

1. How many types of access specifiers are provided in OOP (C++)?

a) 1

b) 2

c) 3

d) 4

Answer: c

2. Which among the following can be used together in a single class?

a) Only private

b) Private and Protected together

c) Private and Public together

d) All three together

Answer: d

3. Which among the following can restrict class members to get inherited?

a) Private

b) Protected

c) Public

d) Manual

Answer: a

4. Which access specifier is used when no access specifier is used with a member of class (java)?

a) Private

b) Default

c) Protected

d) Public

Answer: b

5. Which specifier allows a programmer to make the private members which can be inherited?

a) Private

b) Default

c) Protected

d) Protected and default

Answer: c

6. Which among the following is false?

a) Private members can be accessed using friend functions

b) Member functions can be made private

c) Default members can’t be inherited

d) Public members are accessible from other classes also

Answer: c

7. If a class has all the private members, which specifier will be used for its implicit constructor?

a) Private

b) Public

c) Protected

d) Default

Answer: b

8. If class A has add() function with protected access, and few other members in public. Then class B inherits class A privately. Will the user will not be able to call _________ from the object of class B.

a) Any function of class A

b) The add() function of class A

c) Any member of class A

d) Private, protected and public members of class A

Answer: d

9. Which access specifier should be used in a class where the instances can’t be created?

a) Private default constructor

b) All private constructors

c) Only default constructor to be public

d) Only default constructor to be protected

Answer: b

10. On which specifier’s data, does the size of a class’s object depend?

a) All the data members are added

b) Only private members are added

c) Only public members are added

d) Only default data members are added

Answer: a

11. If class B inherits class A privately. And class B has a friend function. Will the friend function be able to access the private member of class A?

a) Yes, because friend function can access all the members

b) Yes, because friend function is of class B

c) No, because friend function can only access private members of friend class

d) No, because friend function can access private member of class A also

Answer: c

12. If an abstract class has all the private members, then _________

a) No class will be able to implement members of abstract class

b) Only single inheritance class can implement its members

c) Only other enclosing classes will be able to implement those members

d) No class will be able to access those members but can implement.

Answer: a

13. Which access specifier should be used so that all the parent class members can be inherited and accessed from outside the class?

a) Private

b) Default or public

c) Protected or private

d) Public

Answer: d

14. Which access specifier is usually used for data members of a class?

a) Private

b) Default

c) Protected

d) Public

Answer: a

15. Which specifier should be used for member functions of a class?

a) Private

b) Default

c) Protected

d) Public

Answer: a

Array of Objects

1. What is an array of objects?

a) An array of instances of class represented by single name

b) An array of instances of class represented by more than one name

c) An array of instances which have more than 2 instances

d) An array of instances which have different types

Answer: a

2. Which among the following is a mandatory condition for array of objects?

a) All the objects should be of different class

b) All the objects should be of same program classes

c) All the objects should be of same class

d) All the objects should have different data

Answer: c

3. What is the type of elements of array of objects?

a) Class

b) Void

c) String

d) Null

Answer: a

4. If array of objects is declared as given below, which is the limitation on objects?

Class_name arrayName[size];

a) The objects will have same values

b) The objects will not be initialized individually

c) The objects can never be initialized

d) The objects will have same data

Answer: b

5. Which is the condition that must be followed if the array of objects is declared without initialization, only with size of array?

a) The class should have separate constructor for each object

b) The class must have no constructors

c) The class should not have any member function

d) The class must have a default or zero argument constructor

Answer: d

6. When are the array of objects without any initialization useful?

a) When object data is not required just after the declaration

b) When initialization of object data is to be made by the compiler

c) When object data doesn’t matter in the program

d) When the object should contain garbage data

Answer: a

7. If constructor arguments are passed to objects of array then _____________ if the constructors are overloaded.

a) It is mandatory to pass same number of arguments to all the objects

b) It is mandatory to pass same type of arguments to all the objects

c) It is not mandatory to call same constructor for all the objects

d) It is mandatory to call same constructor for all the constructors

Answer: c

8. How the objects of array can be denoted?

a) Indices

b) Name

c) Random numbers

d) Alphabets

Answer: a

9. The objects in an object array _________________________

a) Can be created without use of constructor

b) Can be created without calling default constructor

c) Can't be created with use of constructor

d) Can't be created without calling default constructor

Answer: b

10. The Object array is created in _______________________

a) Heap memory

b) Stack memory

c) HDD

d) ROM

Answer: a

11. If an array of objects is of size 10 and a data value have to be retrieved from 5th object then _________________ syntax should be used.

a) Array_Name[4].data_variable_name;

b) Data_Type Array_Name[4].data_variable_name;

c) Array_Name[4].data_variable_name.value;

d) Array_Name[4].data_variable_name(value);

Answer: a

12. From which index does the array of objects start?

a) 0

b) 1

c) 2

d) 3

Answer: a

Objects

1. Where does the object is created?

a) class

b) constructor

c) destructor

d) attributes

Answer: a

2. How to access the object in the class?

a) scope resolution operator

b) ternary operator

c) direct member access operator

d) resolution operator

Answer: c

3. . How many objects can present in a single class?

a) 1

b) 2

c) 3

d) as many as possible

Answer: d

4. Pick out the other definition of objects.

a) member of the class

b) associate of the class

c) attribute of the class

d) instance of the class

Answer: d

5. Which of these following members are not accessed by using direct member access operator?

a) public

b) private

c) protected

d) both private & protected

Answer: d

Pointers

1. What does the following statement mean?

```
int (*fp) (char*);
```

a) pointer to a pointer

b) pointer to an array of chars

c) pointer to function taking a char* argument and returns an int

d) function taking a char* argument and returning a pointer to int

Answer: c

2. The operator used for dereferencing or indirection is ____

a) *

b) &

c) ->

d) –>>

Answer: a

3. Choose the right option.

string* x, y;

a) x is a pointer to a string, y is a string

b) y is a pointer to a string, x is a string

c) both x and y are pointers to string types

d) y is a pointer to a string

Answer: a

4. Which one of the following is not a possible state for a pointer.

a) hold the address of the specific object

b) point one past the end of an object

c) zero

d) point to a type

Answer: d

5. Which of the following is illegal?

a) int *ip;

b) string s, *sp = 0;

c) int i; double* dp = &i;

d) int *pi = 0;

Answer: c

6. What will happen in the following C++ code snippet?

int a =100, b =200;

int *p = &a, *q = &b ;

p = q

a) b is assigned to a

b) p now points to b

c) a is assigned to b

d) q now points to a

Answer: b